Jadon Zhu

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SUMMARY: Empathetic engineer and award-winning game developer who rapidly ideates and iterates on tools and prototypes to make users' and engineers' lives easier and more fun. Experienced in software engineering, mathematical modeling and optimization, and AI tooling. Ranked Riot Games player with a firsthand understanding of competitive game systems.

EDUCATION

University of California, Los Angeles (UCLA) **Los Angeles, CA**
Bachelor of Science, Applied Mathematics *Expected June 2027*
Minor, Data Science Engineering

- **GPA:** 3.9
- **Coursework**: Machine Learning, Software Construction, Algorithms and Complexity, Statistics and Probability, Computer Science, Linear Algebra, Optimization, Mathematical Modeling

SKILLS AND TOOLS

Programming Languages: Python, C, C++, Go, C#, JavaScript, GDScript
Machine Learning: PyTorch, Optuna, LLM agents, prompt engineering, ablation design, local model hosting
Cloud & Infrastructure: AWS (S3, Lambda, Batch, DynamoDB, GovCloud), Docker, Linux, Git
Game Development: Godot 4, Unity

EXPERIENCE

Machine Learning Engineer Intern **Los Angeles, CA**
Silvus Technologies, a Motorola Solutions Company *Jun 2026 - Sep 2026*

- Built a data-driven, on-device model management system for software-defined radios, pairing a **React** GUI with a **Python** API service, letting engineers hot-swap live classifier learning models in under **30 seconds**.
- Migrated the model training and evaluation pipeline from manual local runs to **AWS GovCloud**, scaling researchers' usable dataset from a single-machine's ~64 GB memory limits to unbounded cloud compute.
- Implemented event-driven ingestion with **S3** notifications and **Lambda**, converting **MATLAB** recordings to **Zarr** and **Parquet** and indexing signal metadata in **DynamoDB**, which cut dataset load time for evaluation jobs from minutes to seconds through chunked reads instead of full-file loads.
- Structured data on a **medallion lakehouse** architecture and moved simulation and evaluation onto **AWS Batch**, giving the team versioned, reproducible dataset snapshots and **3+** parallel evaluation runs where previously one job ran at a time.

Research Assistant (Optimization) **Los Angeles, CA**
UCLA Health *Mar 2026 - Jun 2026*

- Built a constrained-optimization matching system using **Python**, placing 89 mentees across 130 slots with 71 volunteer mentors for the International Health Economics Association's 2026-27 mentoring cohort
- Quantified an **equity-versus-efficiency** trade-off and presented both options, guaranteeing every volunteer mentor at least one mentee cost 0.7% of total match quality and used all 71 mentors instead of 55.
- Wrote self-checks that halted the run on duplicate matches, double-booked slots or excluded pairs, plus **fuzzy matching** over messy free-text data tuned to over-flag rather than miss a conflict.
- Delivered a design-decisions report and a technical guide documenting every judgment call for successors.

Undergraduate Researcher (Machine Learning) **Los Angeles, CA**
Holden Wu Lab, University of California, Los Angeles *Dec 2025 - March 2026*

- Trained 3D multi-resolution U-Net models in **PyTorch** using remote GPUs on 3-D datasets.
- Generated synthetic data using **data augmentation** to expand limited datasets and prevent model overfitting.
- Implemented a diffeomorphic Demons registration pipeline using **SimpleITK** with multi-resolution processing and configurable smoothing parameters.
- Designed an **Optuna**-based **hyperparameter optimization** framework (80 trials) to tune model parameters and optimize composite loss functions.

- Benchmarked **deep learning** inference speed and output quality against classical optimization algorithms like diffeomorphic Demons.

Embedded Firmware Development and Test Engineering Intern **Los Angeles, CA**
Silvus Technologies, a Motorola Solutions Company *Jun 2025 - Aug 2025*

- Engineered an automated documentation pipeline for 300+ JSON-RPC API methods by designing custom **few-shot** instruction prompts for a locally-hosted **LLM** (Qwen2.5), reducing process time by **90%**.
- Developed an input validation program leveraging **concurrent programming** using **Go** to fortify legacy **C** and **C++** API against cybersecurity attacks.
- Configured **Docker** containerization to establish a consistent environment for cross-compilation of embedded firmware from x86 Linux to ARMv7 RISC.
- Conducted integration testing across a **distributed mesh network** of MANET radios, validating JSON-RPC API calls from local machines over Ethernet to wirelessly-connected nodes.

Undergraduate Researcher (Machine Learning) **Los Angeles, CA**
X. William Yang Lab at UCLA Semel Institute for Neuroscience and Human Behavior *Nov 2023 - Jan 2025*

- Trained a convolutional neural network (CNN) using **PyTorch** and transfer learning to segment 3-D images of mouse brain neurons.
- Developed **Python** scripts to automate 3-D image processing in ImageJ, accelerating preprocessing speed by **3x**.
- Performed manual digital neuron reconstruction using Neutube to map out 3-D structures.

PROJECTS AND COMPETITIONS

LLM Autobattler: Agent Evaluation Harness & Prompt Ablation Study **Los Angeles, CA**
Independent Project *Mar 2026 - Present*

- Prototyped a Teamfight Tactics-inspired autobattler in **Godot 4** where an **LLM-powered agent** competes against scripted opponents, designed as both a playable game **simulation** and a controlled AI **evaluation harness**.
- Built a **context engineering** pipeline with board state serialization (ASCII grid), full game replay history, and structured response parsing to maximize agent decision quality across multi-turn strategic reasoning.
- Designed and ran a **2^3 full-factorial prompt ablation study** evaluating 8 configurations (instructions x examples x reflection) with parallel headless experiment orchestration and automated checkpointing.
- Implemented a **reflection mechanism** allowing the agent to exhibit **emergent** gameplay by learning from past mistakes and dynamically crafting alternative, effective strategies.
- Engineered a configurable agent API across **multiple LLM providers** (Claude, ChatGPT).
- Authored a formal technical report detailing experimental design, results, and analysis of how prompt design impacts AI gameplay behavior.

Game Development (Software Engineering) **Los Angeles, CA**
UCLA Fiat Ludum Game Jam *Apr 2025 - Present*
Association for Computing Machinery (ACM) Studio

- Won over **$6000** in prize value, including **1st place** out of 20 teams as a part of the UCLA Fiat Ludum Jam.
- Developed various complex, unique web-based games using **Unity C#** and **Godot** as a lead software engineer working in a **cross-functional** team.
- Leveraged **composition**, **inheritance**, and other object-oriented techniques and SOLID code principles.

Cloud-Hosted Discord Bot **Los Angeles, CA**
UCLA Student Club *Aug 2025 - Present*

- Leveraged **cloud computing** through Oracle Cloud Infrastructure (OCI) to deploy and host custom Discord bots.
- Ensured continuous application uptime and reliability by managing the deployment with PM2 and implementing rotating logs to monitor errors and troubleshoot crashes.